Notation : the notation $a \lesssim b$ means that there exists an absolute constant $C > 0$ such that $a \leq C\, b$.

Sub-Gaussian Distributions

Exercise 1 (Tails, moments and MGF of the normal distribution ☕☕)**.** For $X \sim N(0,1)$, prove the following.

1. We have the tail bound : $\mathbb{P}\{|X| \geq t\} \leq 2e^{-t^2/2}$ for all $t \geq 0$.
2. For each $p \geq 1$, the r.v. X satisfies

$$\|X\|_{L^p} = (E\,|X|^p)^{1/p} = \sqrt{2}\left[\frac{\Gamma\left((1+p)/2\right)}{\Gamma(1/2)}\right]^{1/p}.$$

   Deduce the following behavior of the moments :

$$\|X\|_{L^p} = O\left(\sqrt{p}\right) \quad \text{as } p \to \infty.$$

3. The MGF satisfies :

$$\mathbb{E}\,\exp\left(\lambda X\right) = e^{\lambda^2/2}$$

   for all $\lambda \in \mathbb{R}$.

Exercise 2 (MGF of X ☕)**.** Let X be a r.v. such that $\mathbb{E}\,X = 0$. We recall the property (v) of Proposition 1 : X is sub-Gaussian if, and only if, its MGF satisfies

$$\mathbb{E}\,\exp(\lambda X) \leq \exp\left(K^2\lambda^2\right) \qquad \text{for all} \quad \lambda \in \mathbb{R}, \tag{1}$$

and some $K > 0$. Show that the condition $\mathbb{E}\,X = 0$ is necessary for property (1) to hold.

Exercise 3 (MGF of X^2 ☕☕)**.** We recall the property (iii) of Proposition 1. A r.v. X is sub-Gaussian if, and only if, the MGF of X^2 satisfies

$$\mathbb{E}\,\exp(\lambda^2 X^2) \leq \exp\left(K^2\lambda^2\right) \qquad \text{for all } \lambda \text{ such that } |\lambda| \leq \frac{1}{K}, \tag{2}$$

and some parameter $K > 0$. Show the following.

1. If $X \sim N(0,1)$, the function $\lambda \mapsto \mathbb{E}\,\exp(\lambda^2 X^2)$ is finite only in some bounded neighborhood of zero.
2. If $\mathbb{E}\,\exp(\lambda^2 X^2) \leq \exp\left(K^2\lambda^2\right)$ for all $\lambda \in \mathbb{R}$ and some constant K, then X is bounded, i.e. $\|X\|_\infty < \infty$.

Exercise 4 (sub-Gaussian norm ☕☕)**.** The sub-Gaussian norm of X, denoted $\|X\|_{\phi_2}$, is defined by

$$\|X\|_{\phi_2} = \inf\left\{t > 0 : \ \mathbb{E}\,\exp\left(X^2/t^2\right) \leq 2\right\}.$$

Check that $\|\cdot\|_{\psi_2}$ is indeed a norm on the space of sub-Gaussian r.v.

Exercise 5 (Outside the class of sub-Gaussian r.v. ☕)**.** Check that the Poisson, exponential, Pareto, and Cauchy distributions are not sub-Gaussian.

Exercise 6 (Maximum of sub-Gaussian ☕☕☕)**.** Let $X_1, X_2, \dots,$ be an infinite sequence of sub-Gaussian r.v. which are not necessarily independent.

1. Show that
$$\mathbb{E} \max_i \frac{|X_i|}{\sqrt{1+\log i}} \ \leq \ C\,K,$$
where $K = \max_i \|X_i\|_{\psi_2}$.
2. Deduce that for every $N \geq 2$ we have
$$\mathbb{E} \max_{i \leq N} |X_i| \ \leq \ CK\sqrt{\log\, N}.$$

Exercise 7 (Lower bound ☕☕)**.** Show that the bound of Exercise 6 is sharp. Let $X_1, X_2, \dots,$ be independent $N(0,1)$ r.v.. Prove that
$$\mathbb{E} \max_{i \leq N} |X_i| \ \ \geq \ \ c\,\sqrt{\log\, N}.$$

Exercise 8 (General Hoeffding's inequality ☕)**.** Recall the general Hoeffding's inequality for sub-Gaussian r.v.. Use it to prove Hoeffding's inequality for bounded r.v. (Ex 1, Sheet 2), possibly with some other absolute constant instead of 2 in the exponent.

Exercise 9 (Khintchine's inequality ☕☕)**.** Let $X_1, \dots, X_n$ be independent sub-Gaussian r.v. with zero means and unit variances, and let $a = (a_1, \dots, a_N) \in \mathbb{R}^N$. Prove that for every $p \in [2, \infty)$ we have
$$\left(\sum_{i=1}^N a_i^2\right)^{1/2} \ \leq \ \|\sum_{i=1}^N a_i X_i\|_{L^p} \ \leq \ C\,K\sqrt{p}\left(\sum_{i=1}^N a_i^2\right)^{1/2}$$
where $K = \max_i \|X_i\|_{\psi_2}$ and C is an absolute constant.

Exercise 10 (Khintchine's inequality for $p = 1$ ☕☕☕)**.** Show that, in the setting of Exercise 9, we have
$$c(K)\left(\sum_{i=1}^N a_i^2\right)^{1/2} \ \leq \ \|\sum_{i=1}^N a_i X_i\|_{L^1} \ \leq \ \left(\sum_{i=1}^N a_i^2\right)^{1/2}.$$
Here $K = \max_i \|X_i\|_{\psi_2}$ and $c(K) > 0$ is a quantity that may depend only on K.

Sub-Exponential Distributions

Exercise 11 (☕☕)**.** Prove the equivalence of properties (i)-(v) in Proposition 4 (Course notes) by modifying the proof of Proposition 1.

Exercise 12 (☕☕☕). More generally, consider the class of distributions whose tail decay is of type $\exp(-ct^\alpha)$ or faster. Here $\alpha = 2$ corresponds to sub-Gaussian distributions and $\alpha = 1$ to sub-exponential distributions. State and prove a version of Proposition 4 for such distributions.

Exercise 13 (MGF of $|X|$ ☕). Argue that the bound in property (iii) of Proposition 4 cannot be extended to all λ such that $|\lambda| \leq 1/K_3$.

Exercise 14 (Centering ☕). Prove an analog of the centering lemma (Lemma 1) for sub-exponential r.v. X, i.e. $\|X - \mathbb{E}\,X\|_{\psi_1} \leq C\|X\|_{\psi_1}$.

Exercise 15 (Bernstein's inequality ☕). Let $X_1, \ldots, X_n$ be independent mean-zero, sub-exponential r.v.. Give an upper bound of the tails of the sample mean $\frac{1}{N}\sum_{i=1}^N X_i$. Compare Bernstein's inequality with Hoeffding's inequality.